

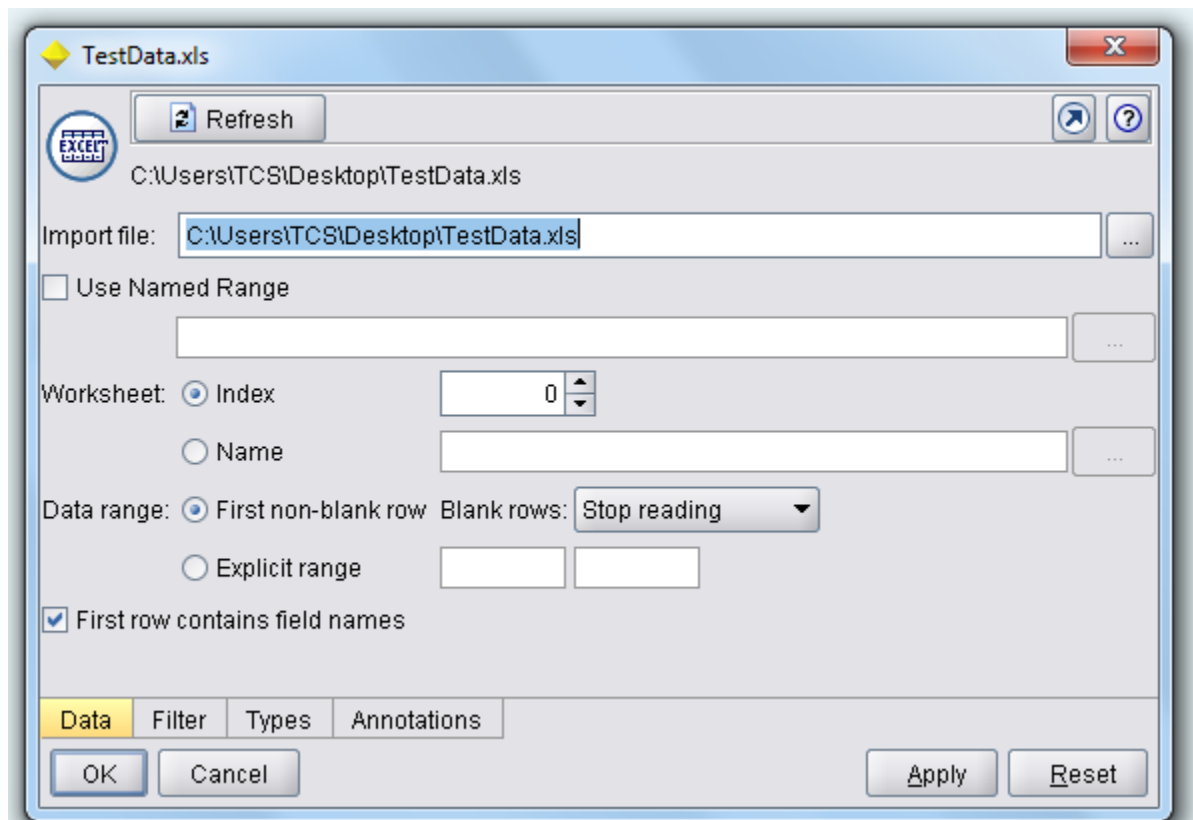
CLEMENTINE TOOL

1. Compare the GRI and Apriori usage (Prepare a sample data set in Spread Sheet). [having support 22% and confidence 90%]

Input data set: An Excel file is prepared as shown below

TID	I1	I2	I3	I4	I5
T100	T	T	F	F	T
T200	F	T	F	T	F
T300	F	T	T	F	F
T400	T	F	T	T	F
T500	T	F	T	F	F
T600	F	T	T	F	F
T700	T	F	T	F	F
T800	T	T	T	F	T
T900	T	T	T	F	F

Procedure: The following are the nodes used for this exercise with respective settings.



Type

Read Values Clear Values Clear All Values

Field	Type	Values	Missing	Check	Direction
TID	Set	T100,T200...		None	None
I1	Flag	T/F		None	Both
I2	Flag	T/F		None	Both
I3	Flag	T/F		None	Both
I4	Flag	T/F		None	Both
I5	Flag	T/F		None	Both

☒ View current fields ☐ View unused field settings

Types Format Annotations

OK Cancel Apply Reset

Settings for Apriori

I1 & I2 & I3 & I4 & I5

Model name: ☒ Auto ☐ Custom

☒ Use partitioned data

Minimum antecedent support (%): 22.0

Minimum rule confidence (%): 90.0

Maximum number of antecedents: 4

☒ Only true values for flags

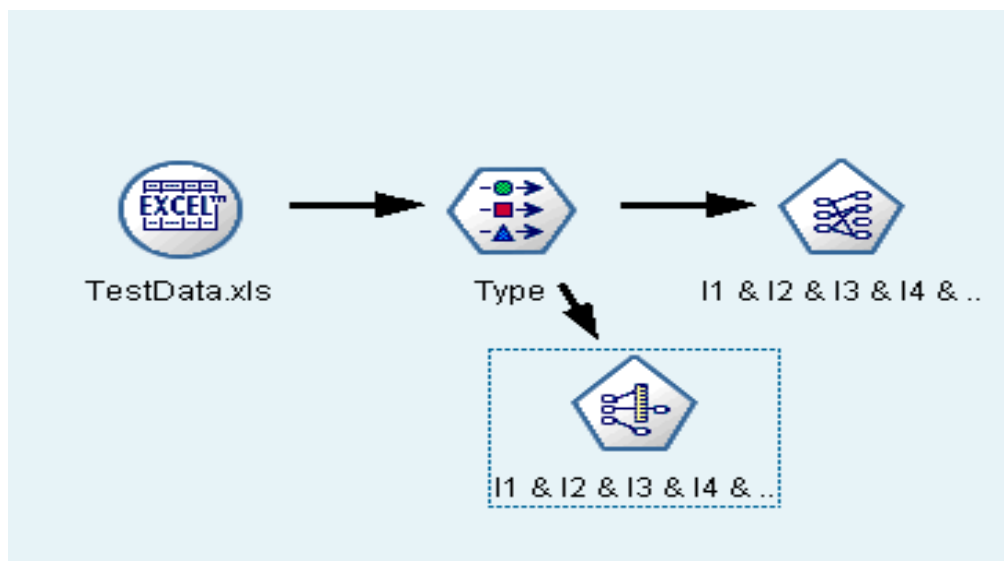
Optimize: ☒ Speed ☐ Memory

Fields Model Expert Annotations

OK Execute Cancel Apply Reset

OUTPUT:

Consequent	Antecedent	Support %	Confidence %
I2	I5	22.222	100.0
I1	I5	22.222	100.0
I1	I5	22.222	100.0
I2	I5	22.222	100.0

OUTPUT: (PICTORIAL WAY)

8. Determine the Drugs importance w.r.t. Age, Cholestrol and BP using C 5.0.10. Compare the C 5.0 and Neural Net using the sample data.

Input data set:

DRUG4n

Refresh

C:\Program Files (x86)\SPSS Clementine\11.1\Demos\DRUG4n

File: C:\Program Files (x86)\SPSS Clementine\11.1\Demos\DRUG4n

Age, Sex, BP, Cholesterol, Na, K, Drug
48, F, HIGH, NORMAL, 0.692623, 0.055369, drugA
15, M, NORMAL, HIGH, 0.678247, 0.040851, drugY
37, M, HIGH, NORMAL, 0.538192, 0.06978, drugA

☒ Read field names from file ☐ Specify number of fields 1

Skip header characters: 0 EOL comment characters:

Strip lead and trail spaces: ☒ None ☐ Left ☐ Right ☐ Both

Invalid characters: ☒ Discard ☐ Replace with

Encoding: Stream default Decimal symbol: Stream default

Lines to scan for type: 50

Delimiters

☐ Space ☒ Comma ☐ Tab
☒ Newline ☐ Other
☐ Non-printing characters
☐ Allow multiple blank delimiters

Quotes

Single quotes: Discard
Double quotes: Discard

File Data Filter Types Annotations

OK Cancel Apply Reset

Procedure: The following are the nodes used for this exercise with respective settings.

Field	Type	Values	Missing	Check	Direction
Age	Range	[15,74]		None	In
Sex	Flag	M/F		None	In
BP	Set	HIGH,LO...		None	In
Cholesterol	Flag	NORMAL/...		None	In
Na	Range	[0.500517,...		None	In
K	Range	[0.020152,...		None	In
Drug	Set	drugA,dru...		None	In

☒ View current fields ☐ View unused field settings

Types Format Annotations

OK Cancel Apply Reset

Settings for C5.0 Model

Model name: ☒ Auto ☐ Custom

☒ Use partitioned data

Output type: ☒ Decision tree ☐ Rule set

☐ Group symbolics

☐ Use boosting Number of trials: 10

☐ Cross-validate Number of folds: 10

Mode: ☒ Simple ☐ Expert

Favor: ☒ Accuracy ☐ Generality

Expected noise (%): 0

Fields Model Costs Annotations

OK Execute Cancel Apply Reset

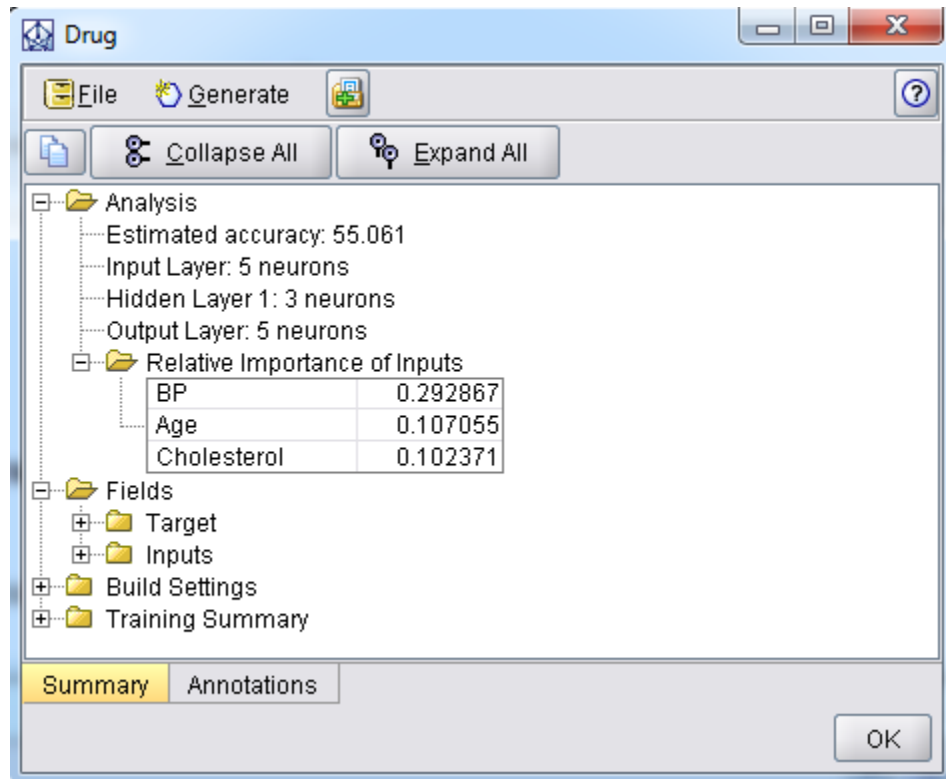
Settings for Neural Net Model

The screenshot shows a software window titled "Drug" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there is a toolbar with a neural network icon and a help icon. The main area contains the following settings:

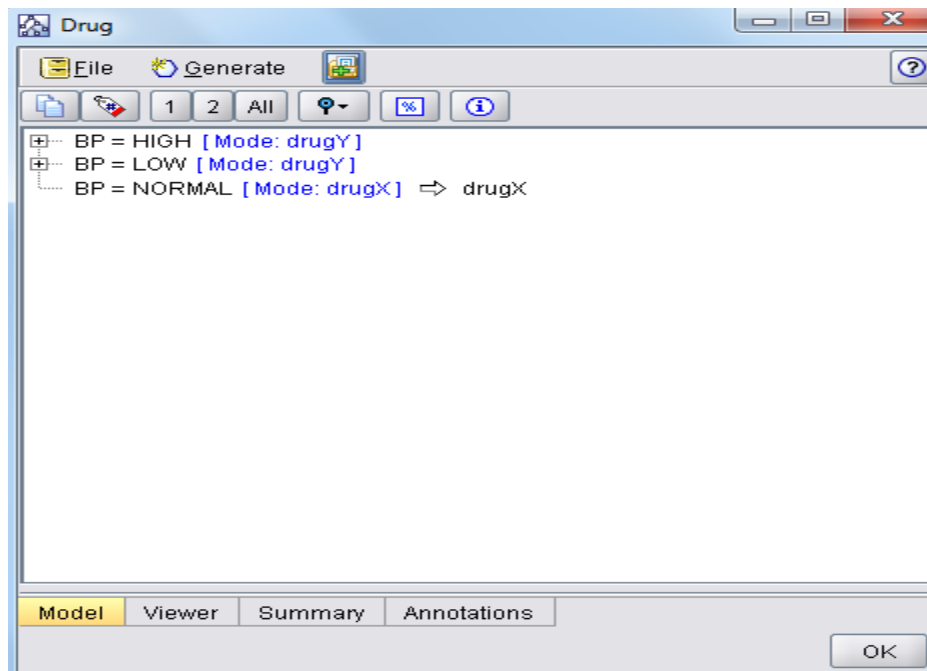
- Model name:** Radio buttons for "Auto" (selected) and "Custom". A text input field is next to the "Custom" option.
- Use partitioned data:** A checked checkbox.
- Method:** A dropdown menu currently showing "Quick".
- Prevent overtraining:** A checked checkbox. To its right is a "Sample %" spinner box set to 50.0.
- Set random seed:** An unchecked checkbox. To its right is a "Seed:" spinner box set to 0.
- Stop on:** Radio buttons for "Default" (selected), "Accuracy (%)" (set to 90.0), "Cycles" (set to 250), and "Time (mins)" (set to 5.0).
- Optimize:** Radio buttons for "Speed" and "Memory" (selected).

At the bottom of the window is a tabbed interface with five tabs: "Fields", "Model" (highlighted in yellow), "Options", "Expert", and "Annotations". Below the tabs are five buttons: "OK", "Execute" (with a green play icon), "Cancel", "Apply", and "Reset".

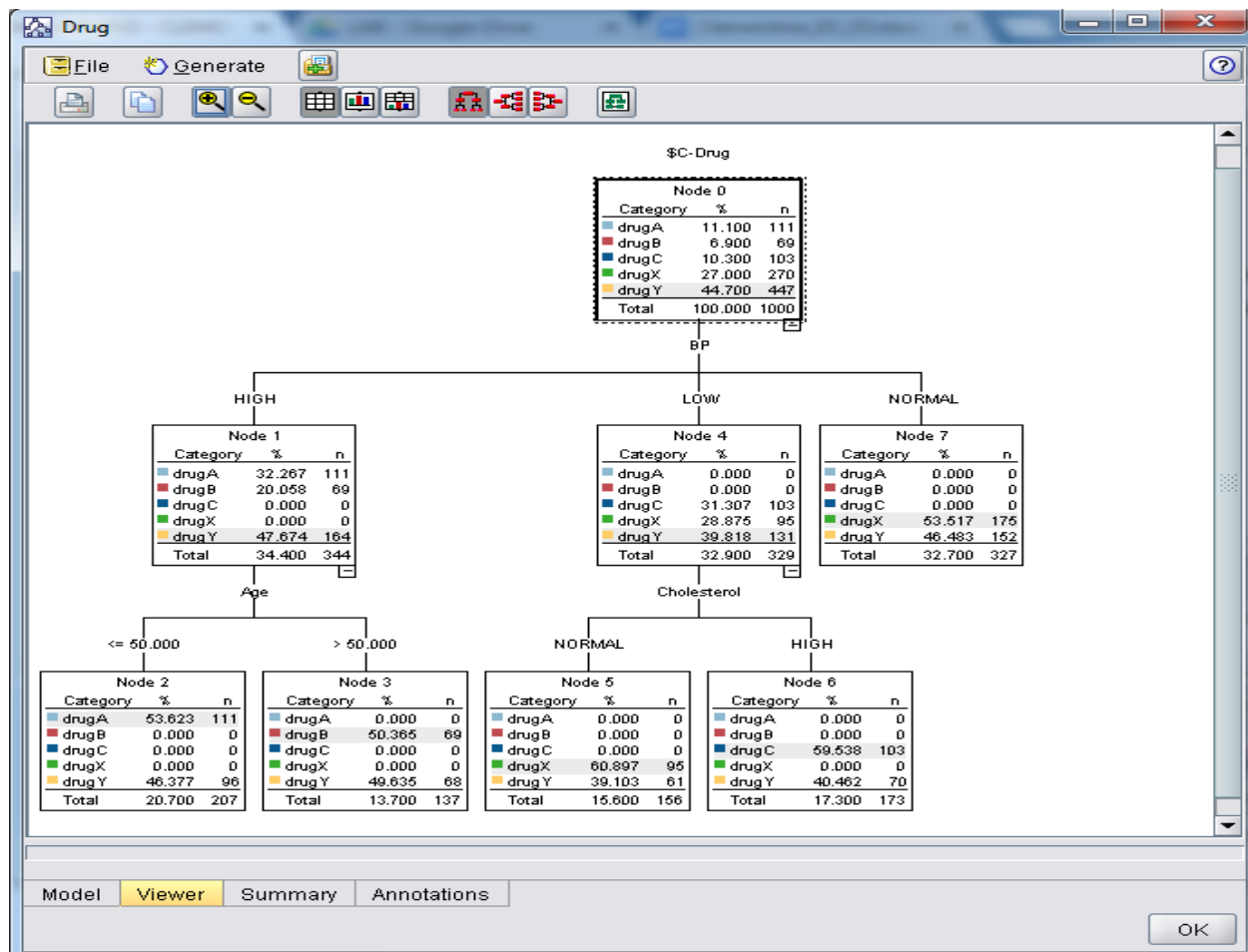
Output of Neural Net Model



Output of C5.0 Model



Output of C5.0 Model viewer which is shown in tree format



Comparison: Attribute Importance is as follows

Neural net Model : BP- 0.2928, Age- 0.1070, Cholestrol- 0.1023

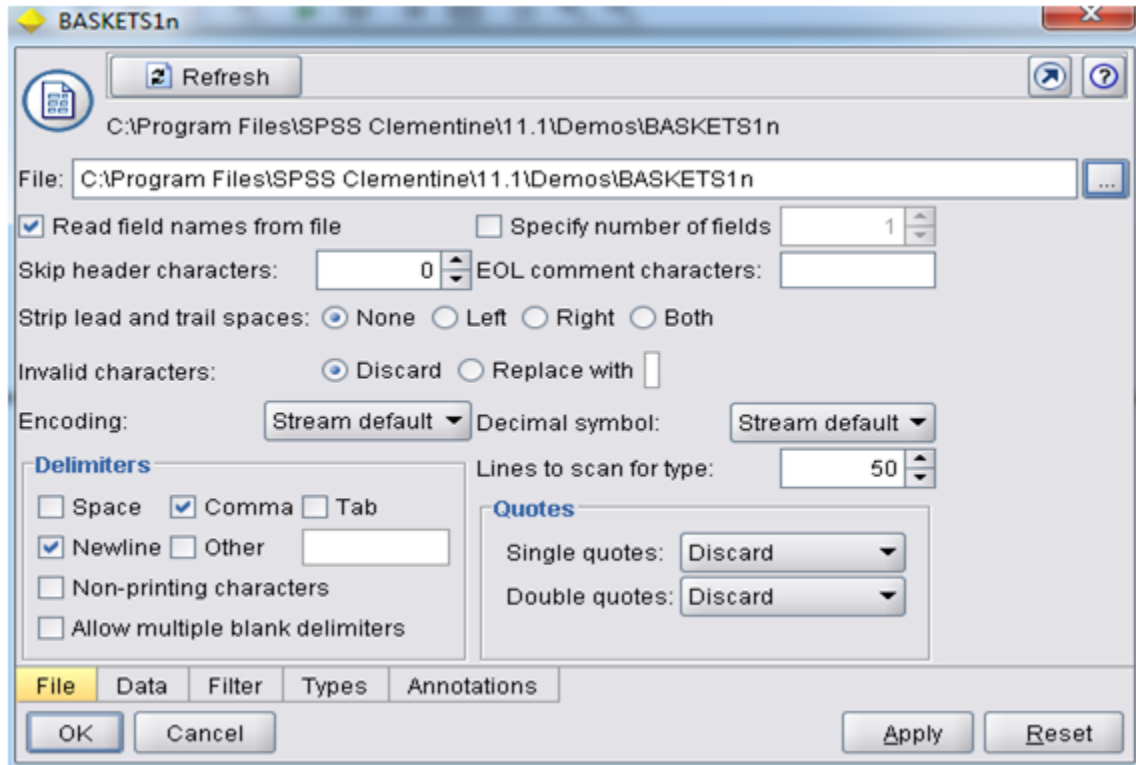
C 5.0 : 2 Level Decision Tree is prepared with BP as root attribute (level 1) with

Age and Cholesterol Attributes at level 2

** As a result Neural Net and C 5.0 Models are giving the same information.

11. Using the BASKETS1n dataset, select the data as given below.

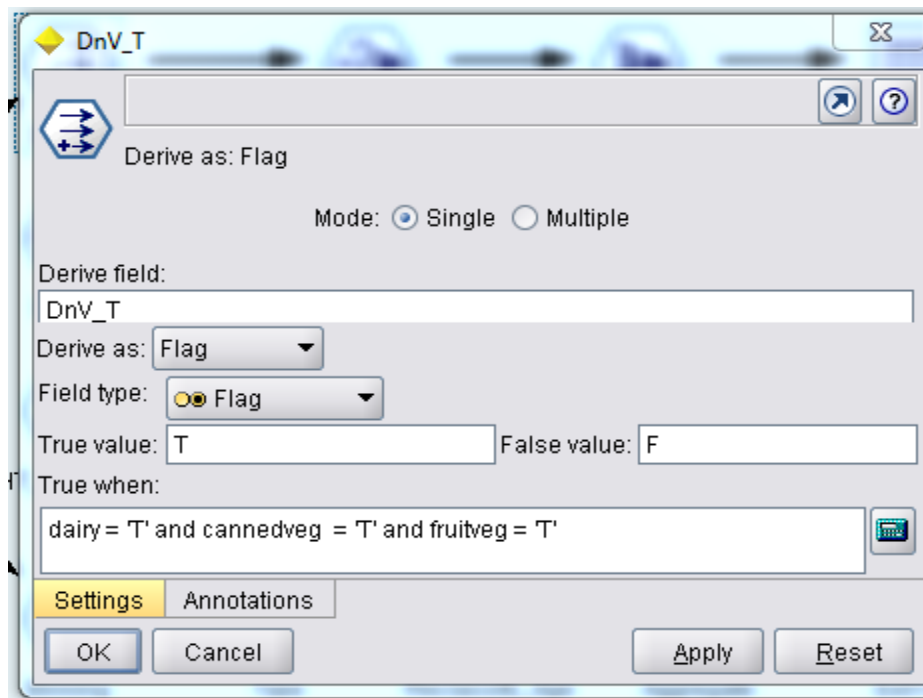
Input data set is applicable to all exercises in given problem statement: BASKETS1n



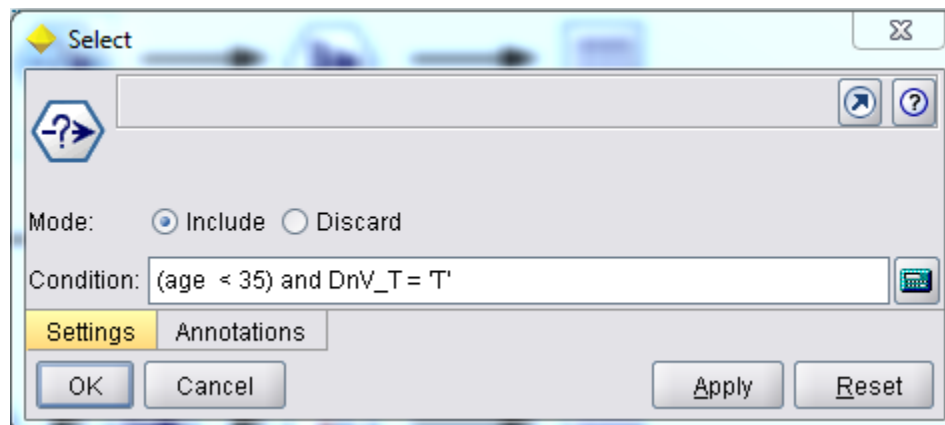
a) Customer age < 35 and count the customers who buy dairy and VEG products

Procedure:

1. Specify the name of the file. You can enter a filename or click the ellipsis button (...) to select a file. The file path is shown once you have selected a file, and its contents are displayed with delimiters in the panel below it.
2. Select var.file from sources then goto C:\Program Files (x86)\SPSS Clementine\11.1\Demos\BASKETS1n we get the baskets in file.
3. Go to field options and select Derive flag and give condition as dairy = 'T' and cannedveg = 'T' and fruitveg = 'T' and click OK based on the conditions the truth values are shown and records are selected.

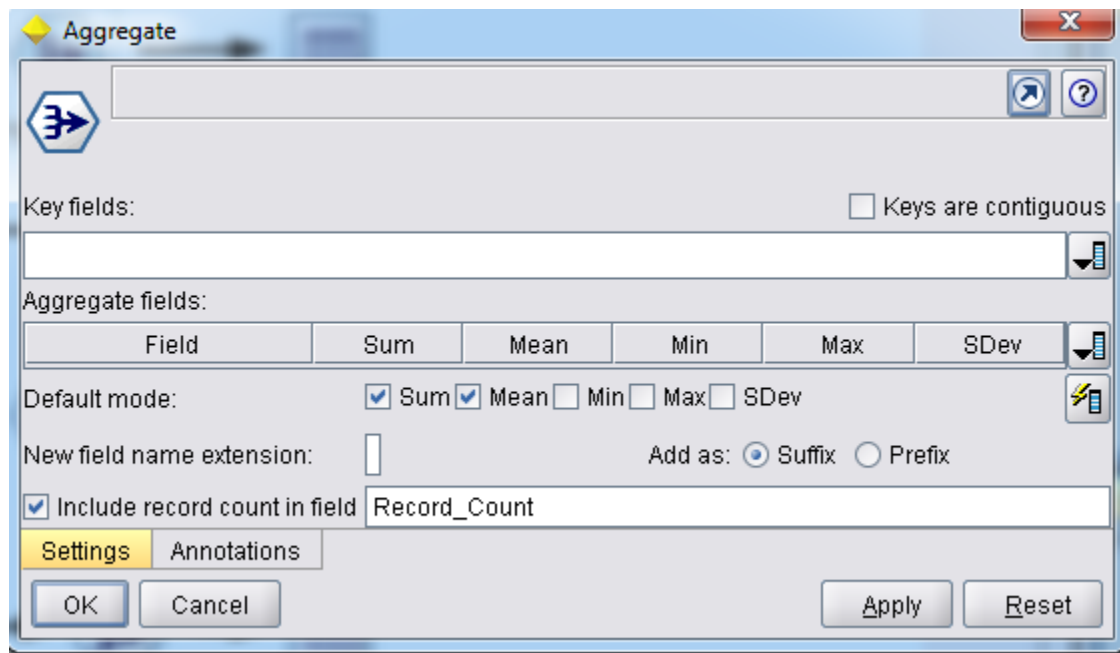


4. Goto options search for the Select give the condition as [(age < 35) and DnV_T = 'T']



5. If both the conditions are true it counts the number of records.

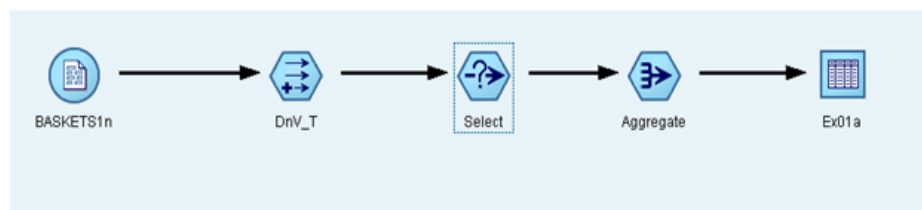
6. Select the Aggregate operation to retrieve sum and max of the records.



7. Select output table and connect with aggregator
8. Right click on output table and execute it.
9. Then click on run now output will be displayed.

OUTPUT:

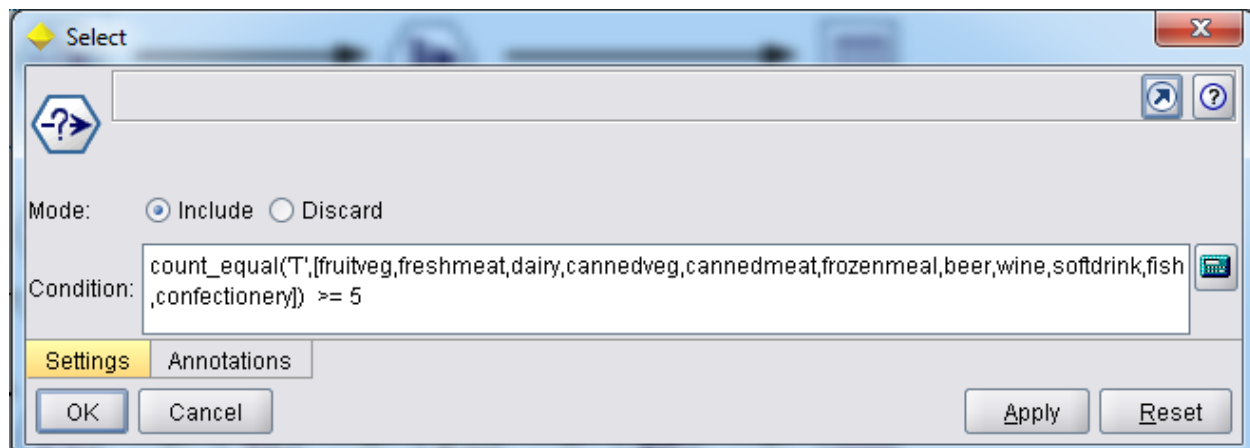
Record_Count
10



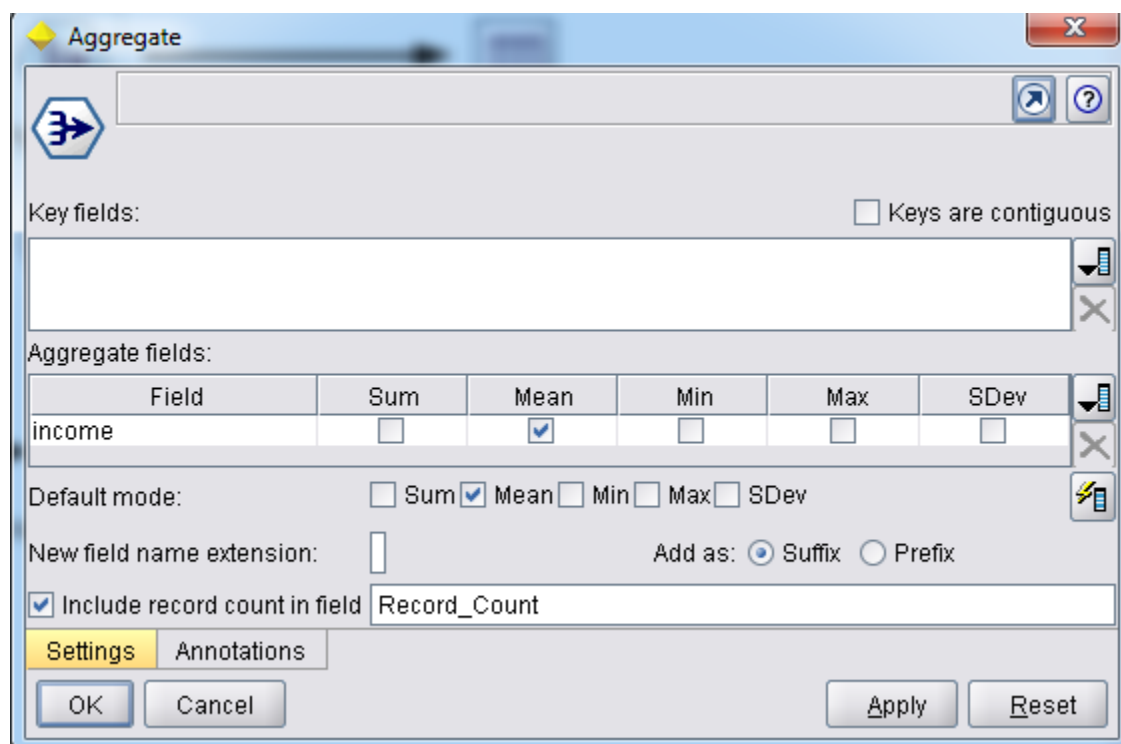
b) Find the AVG income of customers who buy at least 5 products

Procedure:

1. Specify the name of the file. You can enter a filename or click the ellipsis button (...) to select a file. The file path is shown once you have selected a file, and its contents are displayed with delimiters in the panel below it.
2. Select var.file from sources then goto C:\Program Files (x86)\SPSS Clementine\11.1\Demos\BASKETS1n we get the baskets in file.
3. Goto options search for the Select give the condition as shown in figure.

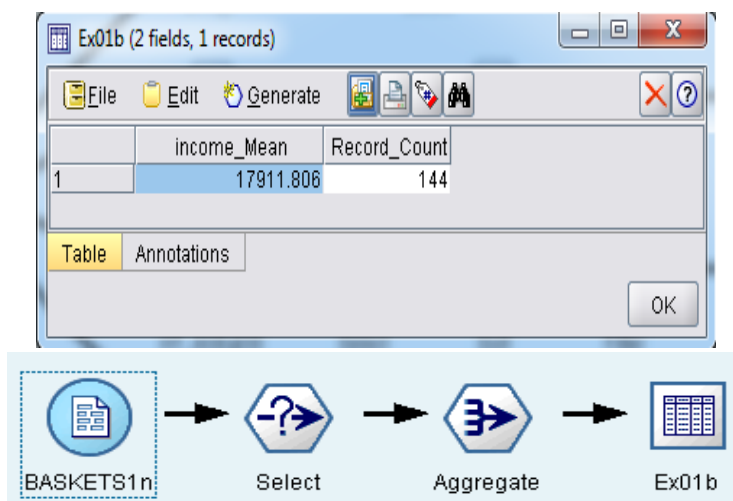


4. If the condition is true it counts the number of records.
5. Select the Aggregate operation to retrieve income_Mean and count of the records.



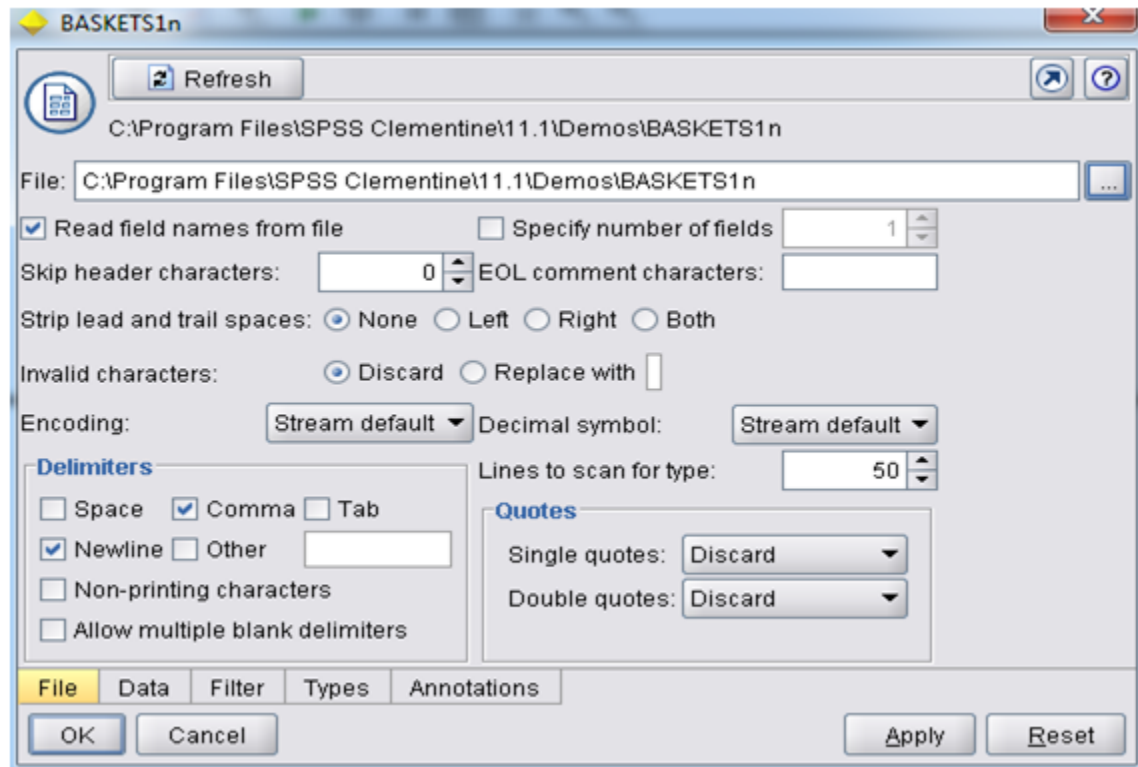
6. Select output table and connect with aggregator
7. Right click on output table and execute it.
8. Then click on run now output will be displayed.

OUTPUT:



12. Using the BASKETS1n dataset, select the data as given below.

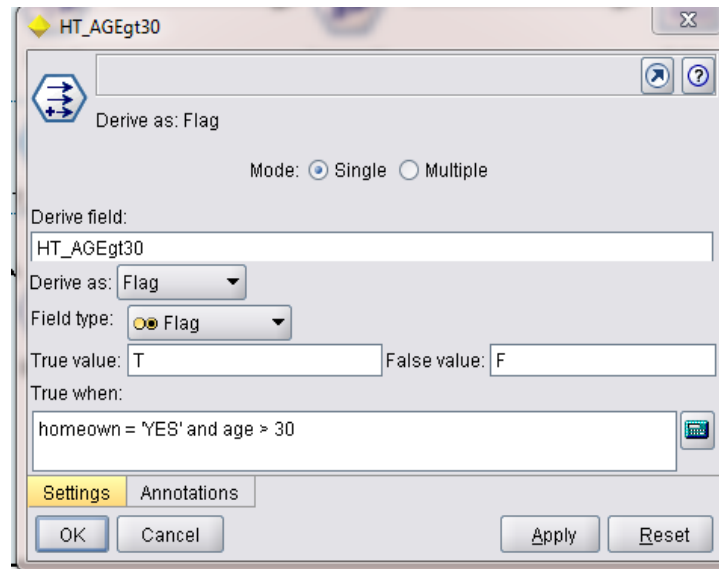
Input data set is applicable to all exercises in given problem statement: BASKETS1n



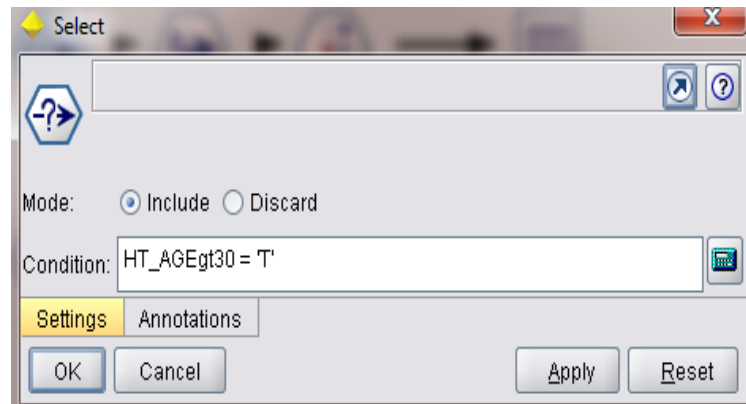
a) Derive the field whose home own is 'YES' and Age > 30 and sort data w.r.t. income in Ascending order, and output only the item fields.

Procedure:

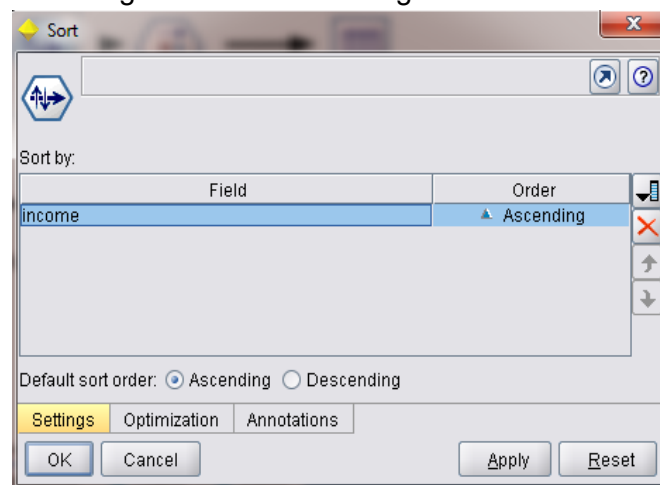
1. Specify the name of the file. You can enter a filename or click the ellipsis button (...) to select a file. The file path is shown once you have selected a file, and its contents are displayed with delimiters in the panel below it.
2. Select var.file from sources then goto C:\Program Files (x86)\SPSS Clementine\11.1\Demos\BASKETS1n we get the baskets in file.
3. Goto Field ops search for the Derive give the condition as shown in figure.



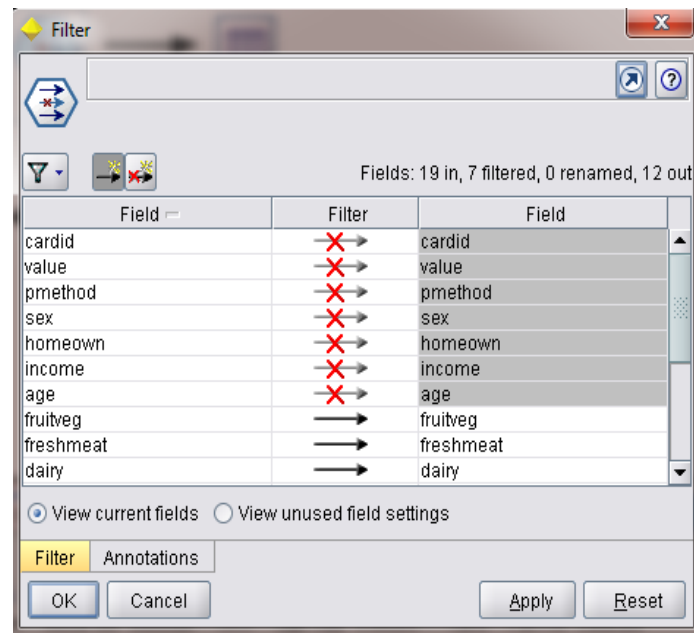
4. Connect Derive to Select with the condition shown in figure



5. Connect Select to Sort for sorting Income in ascending order



6. Connect Sort to Filter out all the non-item fields.



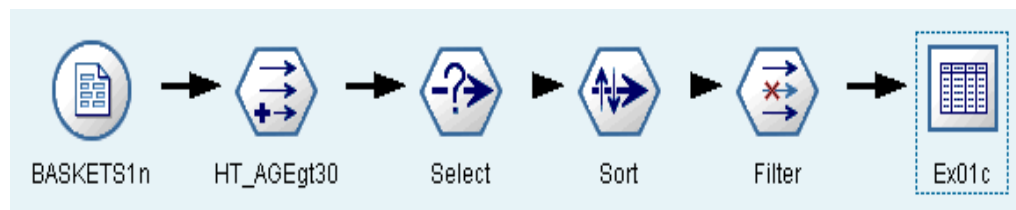
7. Select output table and connect with Filter.

8. Right click on output table and execute it.

9. Then click on run now output will be displayed.

Output dataset:

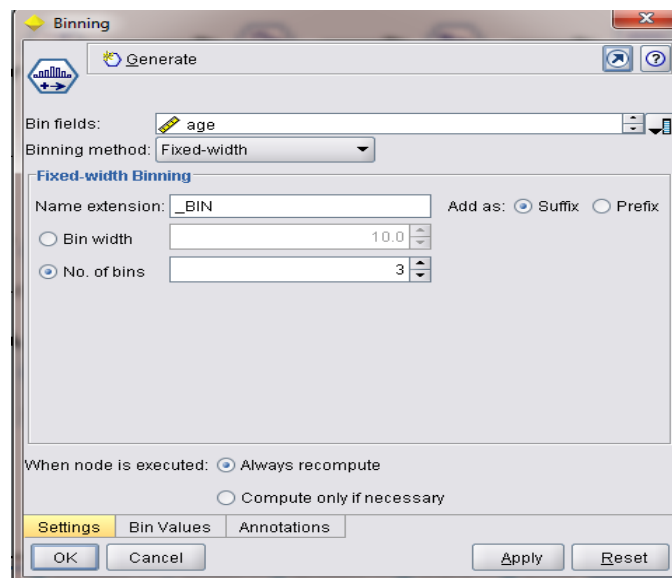
	fruitveg	freshmeat	dairy	cannedveg	cannedmeat	frozenmeal	beer	wine	softdrink	fish	confectionery	HT_AGEgt30
1	F	F	F	T	F	T	T	F	F	F	F	T
2	T	F	F	F	F	F	T	F	F	F	F	T
3	F	F	F	F	F	F	F	F	F	F	T	T
4	F	F	F	T	F	T	T	F	F	F	F	T
5	F	T	F	F	F	F	F	F	F	F	T	T
6	F	T	F	T	F	T	T	F	F	T	F	T



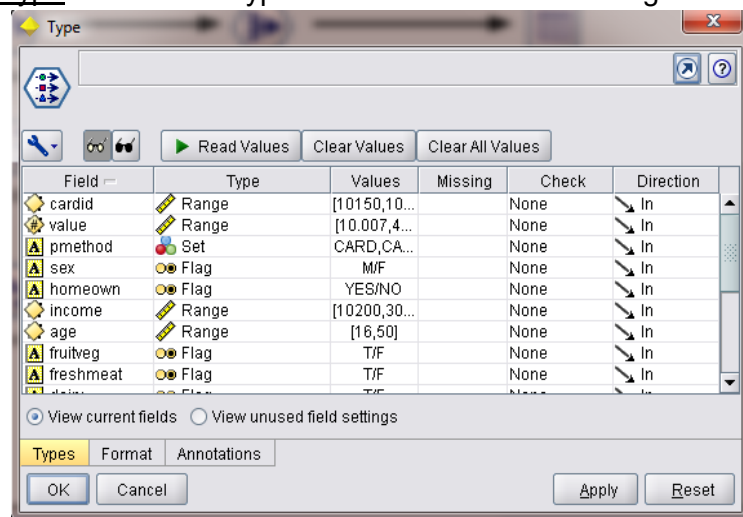
b) Find the mean value of salary w.r.t age = {Young, Middle, Senior}.

Procedure:

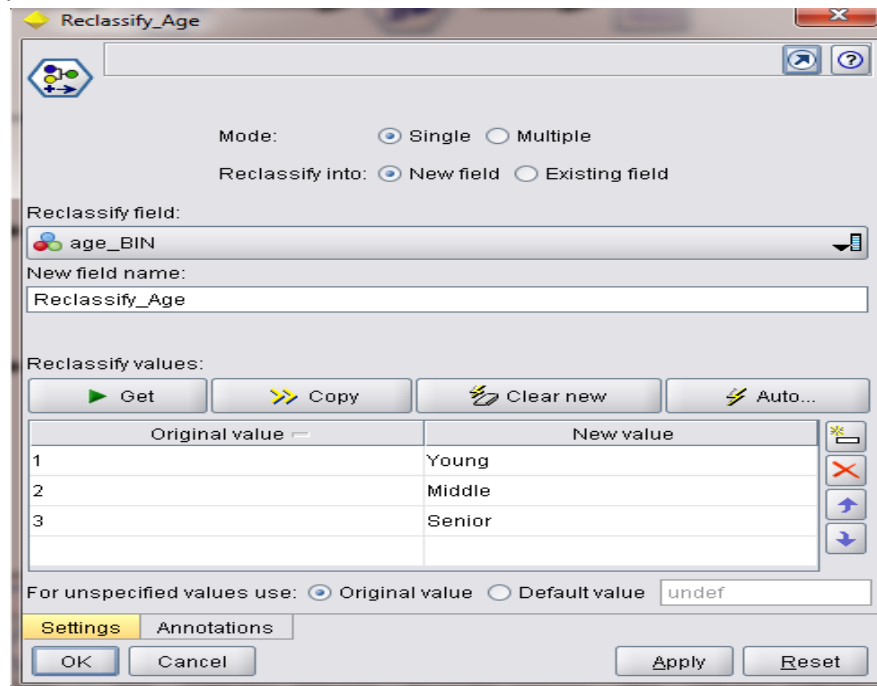
1. Specify the name of the file. You can enter a filename or click the ellipsis button (...) to select a file. The file path is shown once you have selected a file, and its contents are displayed with delimiters in the panel below it.
2. Select var.file from sources then goto C:\Program Files (x86)\SPSS Clementine\11.1\Demos\BASKETS1n we get the baskets in file.
3. Goto Field ops search for the Binning give the condition as as shown in figure.



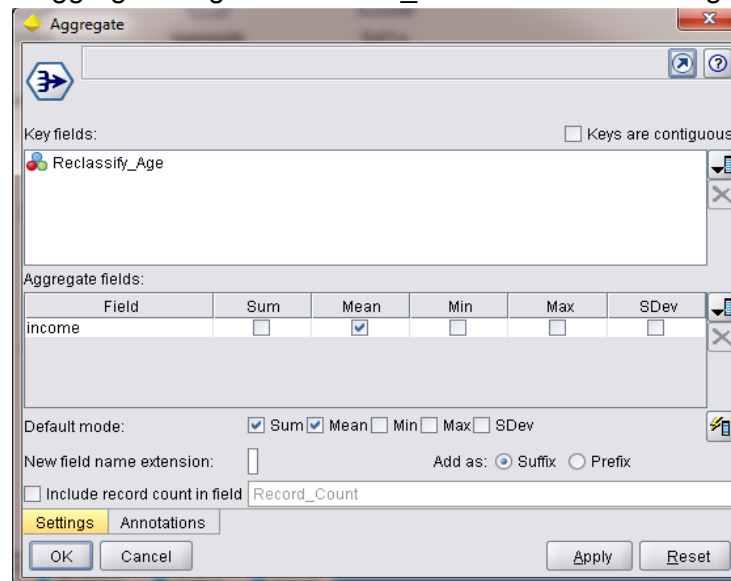
4. Connect Binning to Type to read data types and values as shown in figure



5. Connect Type to Reclassify, for classifying age binned in 1,2,3 to Young,Middle,Senior respectively



6. Connect Reclassify to Aggregate to get the income_Mean w.r.t. Different age categories.



7. Select output table and connect with Aggregate.

8. Right click on output table and execute it.

9. Then click on run now output will be displayed.

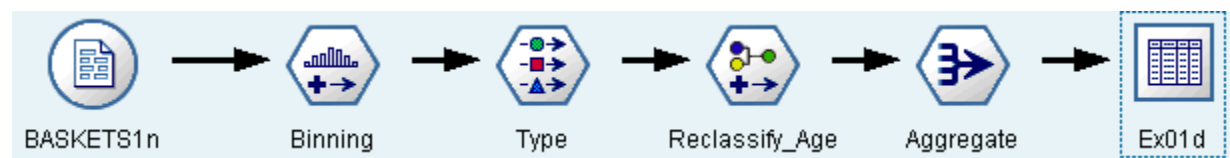
Output dataset:

Ex01d (2 fields, 3 records) #6

	income_Mean	Reclassify_Age
1	19868.852	Young
2	20517.869	Middle
3	20201.458	Senior

Table Annotations

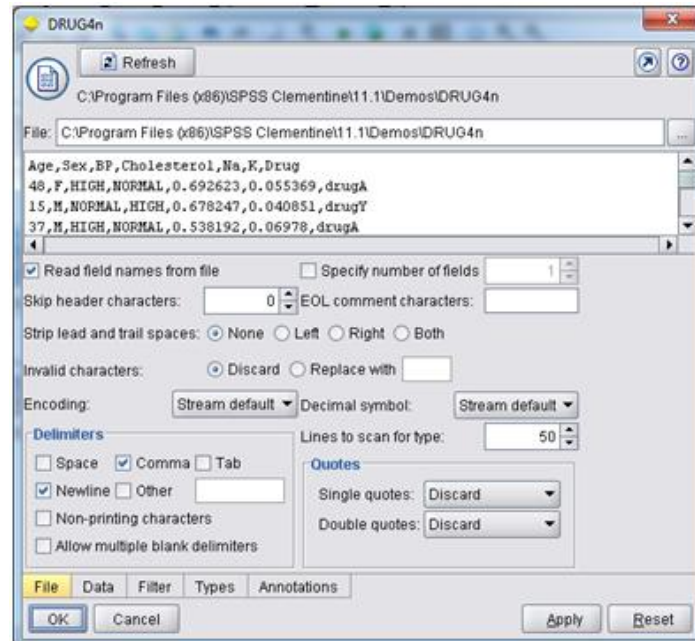
OK



Extra Program

Using DRUG3n and DRUG4n datasets select the data as given below

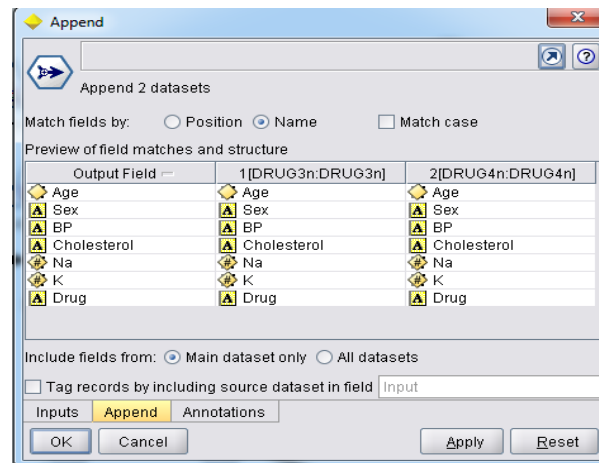
Input data set is applicable to all exercises in given problem statement: DRUG3n and DRUG4n (For excercises d and e DRUG2n is also used)



- a) Select 50% of records where maximum type of drug are present along with no restrictions on remaining drugs, and use histogram graph of age w.r.t BP

Procedure: The following are the nodes used for this exercise with respective settings.

APPEND:



Aggregate: to get the count of all Drug types as shown in following output table.

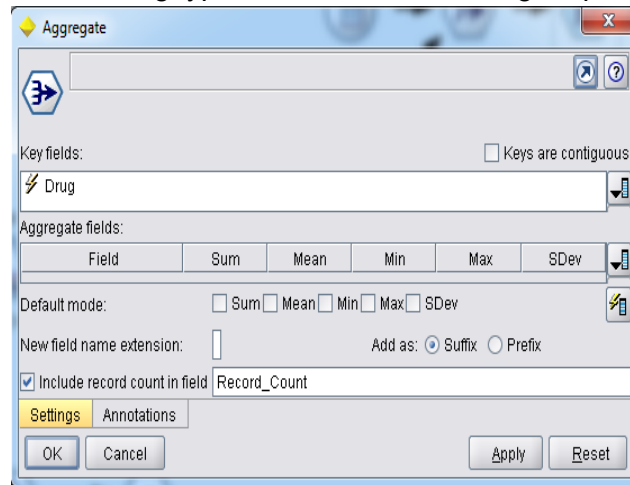


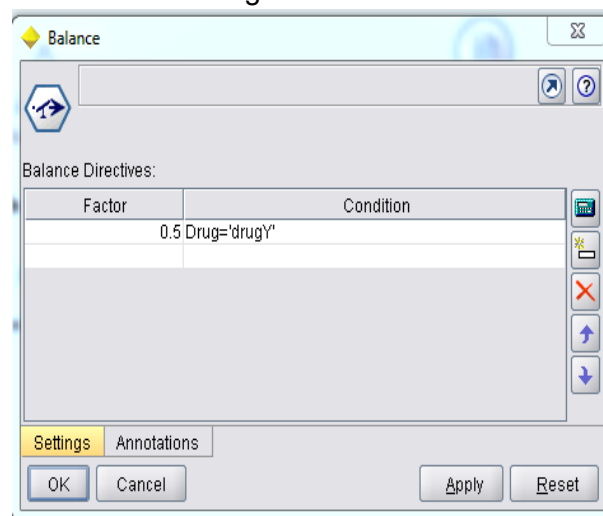
Table: from the following output we can identify that 'drugY' has maximum number of records when compared to remaining Drug types.

Table (2 fields, 5 records) #4

	Drug	Record_Count
1	drugB	121
2	drugX	426
3	drugC	153
4	drugY	719
5	drugA	181

Table Annotations

Balance: for selecting 50% of records for 'drugY'



Histogram: The following Histogram node gives the output of Age w.r.t. BP

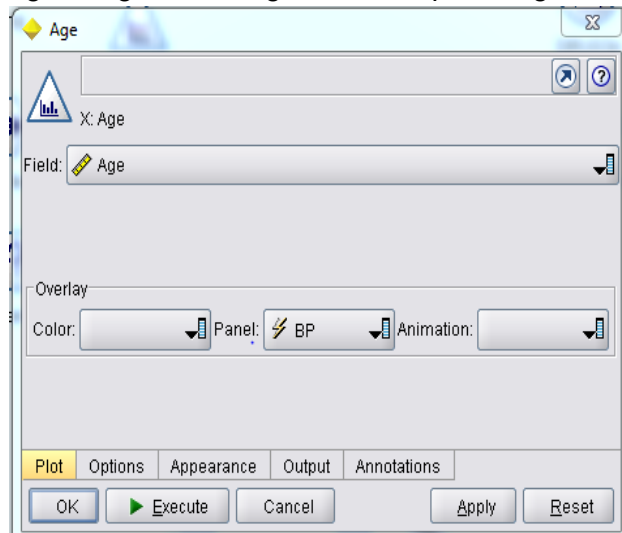
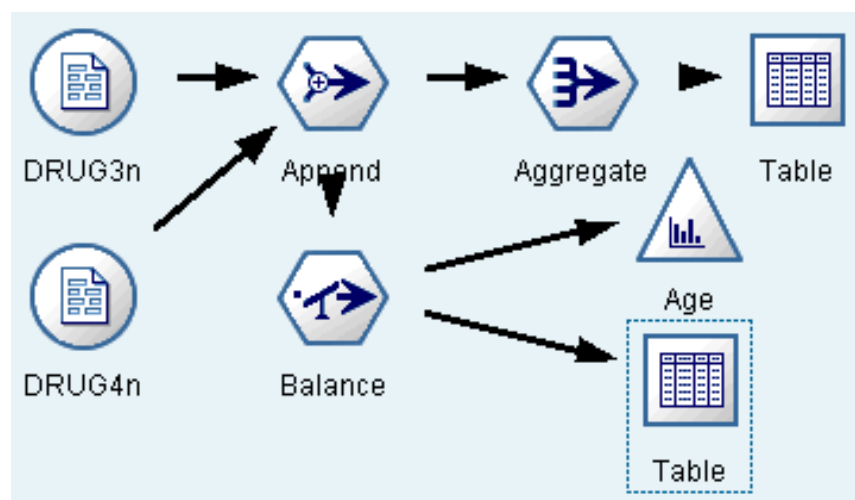
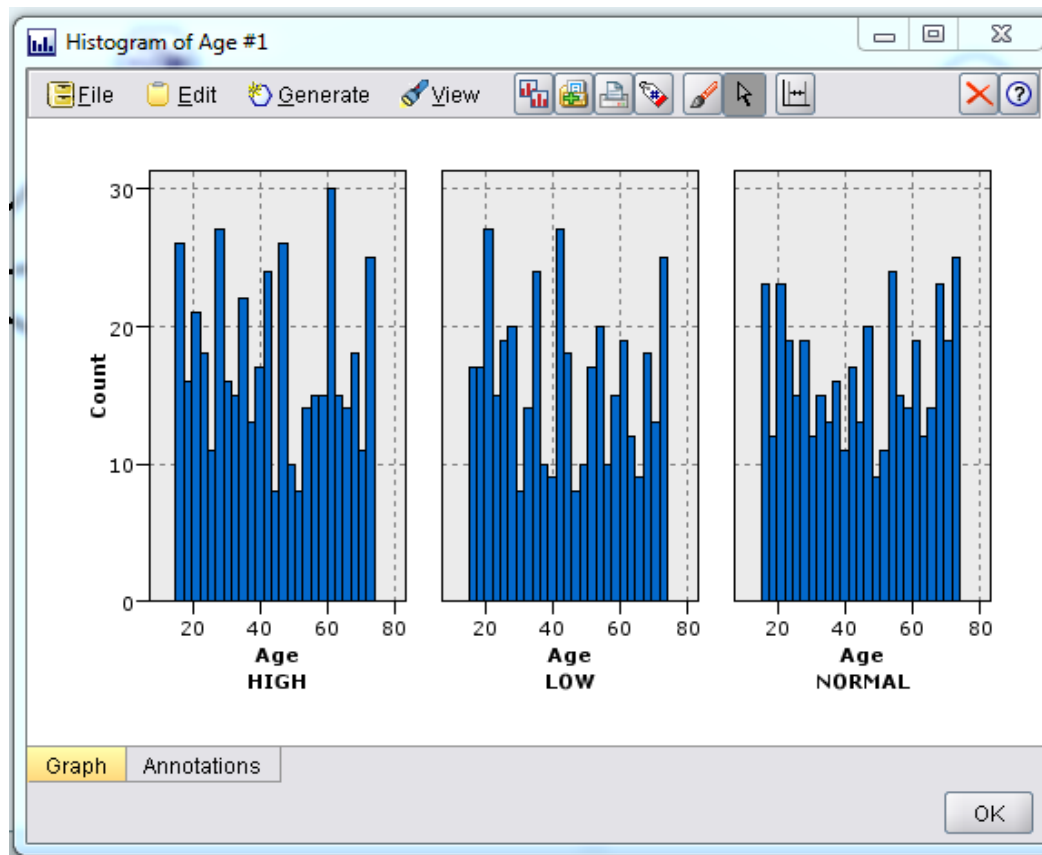


Table: It is an output of the records after selecting 50% of 'drugY' and no restrictions on remaining Drugs

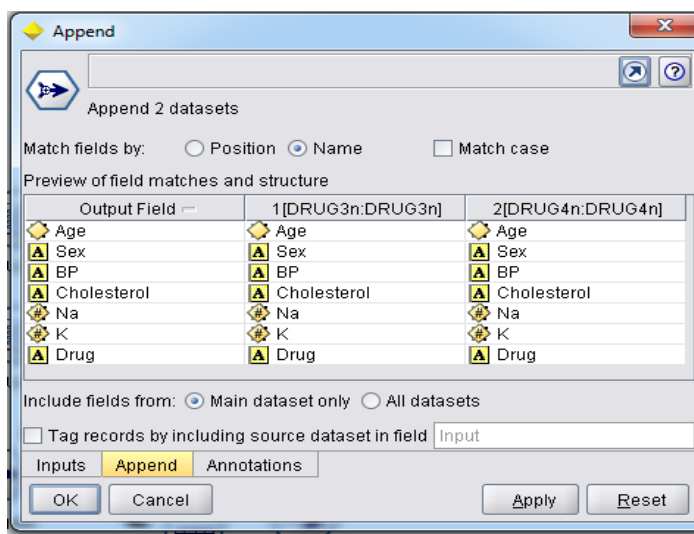
	Age	Sex	BP	Cholesterol	Na	K	Drug
1	25	F	HIGH	HIGH	0.676	0.075	drugA
2	17	F	HIGH	HIGH	0.540	0.030	drugY
3	74	F	LOW	HIGH	0.850	0.077	drugC
4	40	F	NORMAL	HIGH	0.677	0.050	drugX
5	32	F	HIGH	HIGH	0.582	0.025	drugY
6	64	M	HIGH	NORMAL	0.641	0.078	drugB
7	45	M	HIGH	HIGH	0.664	0.048	drugA
8	33	F	LOW	NORMAL	0.822	0.028	drugY
9	73	M	HIGH	NORMAL	0.792	0.062	drugB
10	27	F	HIGH	NORMAL	0.555	0.047	drugA
11	62	M	HIGH	NORMAL	0.510	0.071	drugB
12	72	M	HIGH	NORMAL	0.819	0.074	drugB
13	19	M	HIGH	NORMAL	0.553	0.033	drugY
14	28	M	HIGH	HIGH	0.584	0.068	drugA
15	54	M	NORMAL	NORMAL	0.606	0.077	drugX
16	37	F	NORMAL	HIGH	0.744	0.021	drugY
17	33	F	NORMAL	NORMAL	0.815	0.065	drugX
18	73	F	LOW	HIGH	0.523	0.026	drugY
19	41	F	NORMAL	HIGH	0.797	0.075	drugX
20	41	M	LOW	HIGH	0.846	0.055	drugY

OUTPUT:

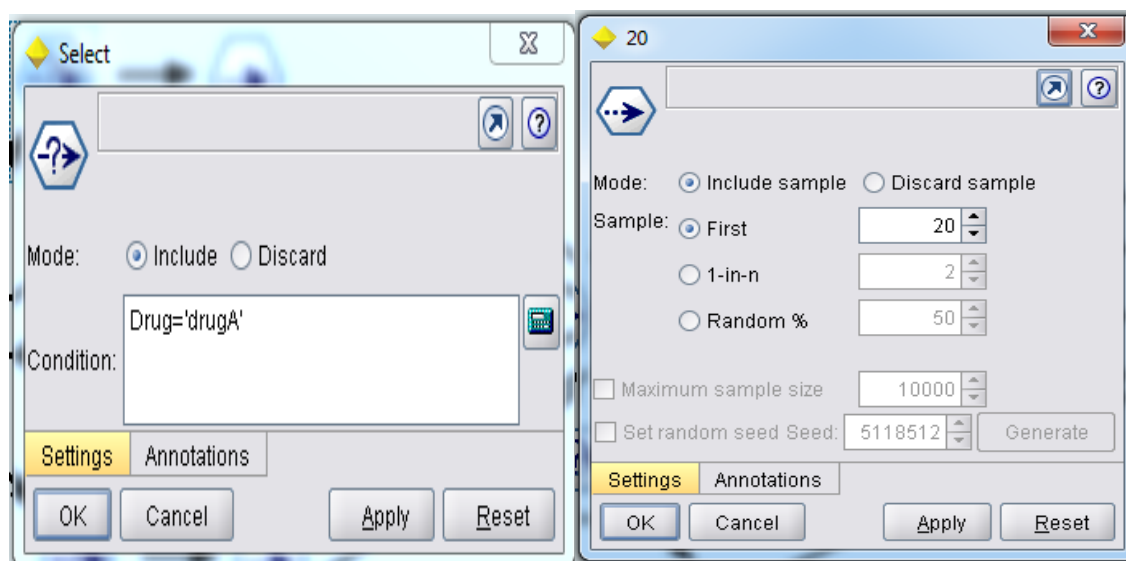
- b) Take the equal number of samples of each drug and calculate the Std. Dev. of age w.r.t drug and compare it with complete data Std. Dev. of age w.r.t drug and give a conclusion statement.

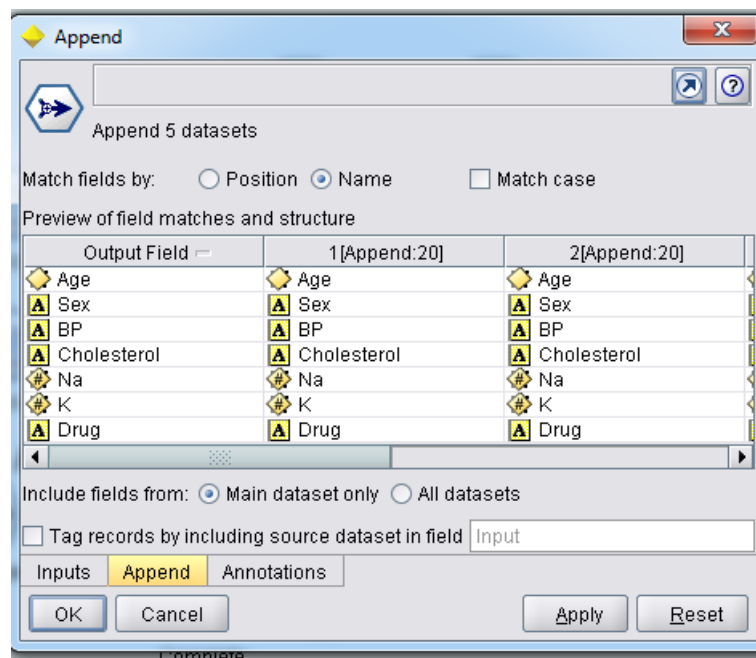
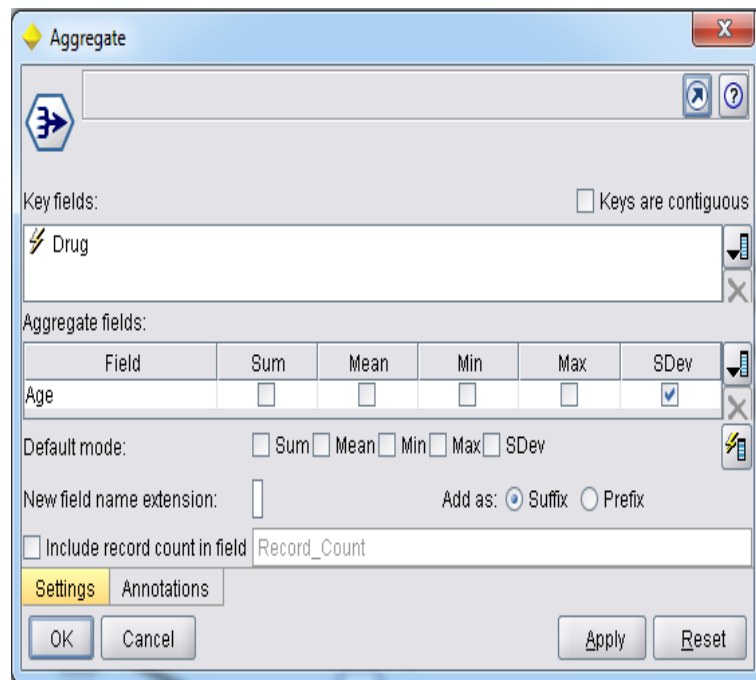
Procedure: The following are the nodes used for this exercise with respective settings.

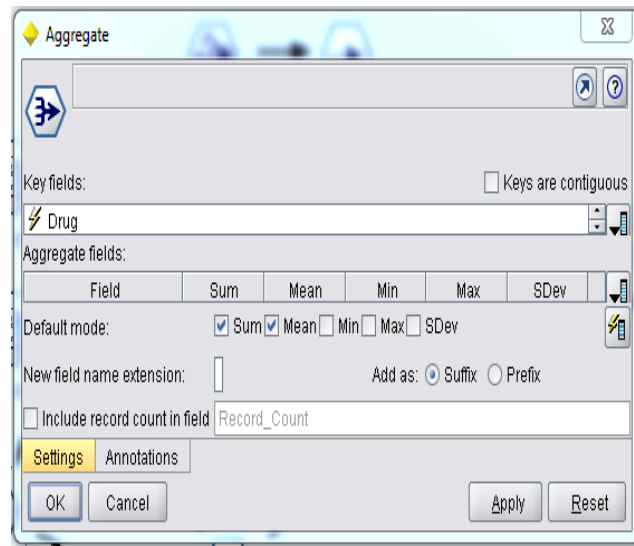
APPEND



SELECT and SAMPLE: This procedure is followed for remaining drug types types where 20 equal samples of each drug type is selected



APPEND: Appending all samples**Sample Aggregate:** Aggregate of Sample data

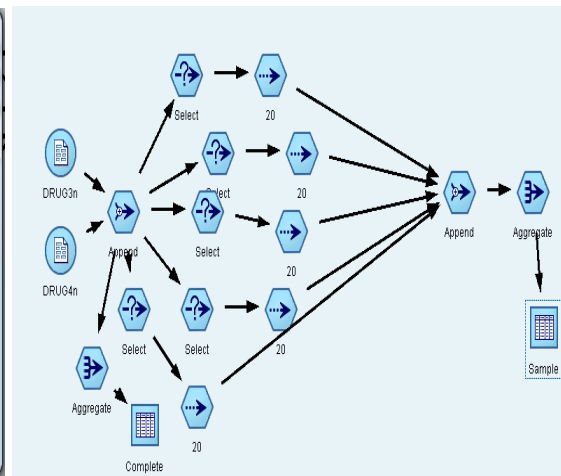
Complete Aggregate: Aggregate of Complete data**OUTPUT:**

Sample (2 fields, 5 records) #1

	Age_SDev	Drug
1	20.072	drugY
2	18.847	drugX
3	17.136	drugC
4	6.307	drugB
5	11.537	drugA

Table Annotations

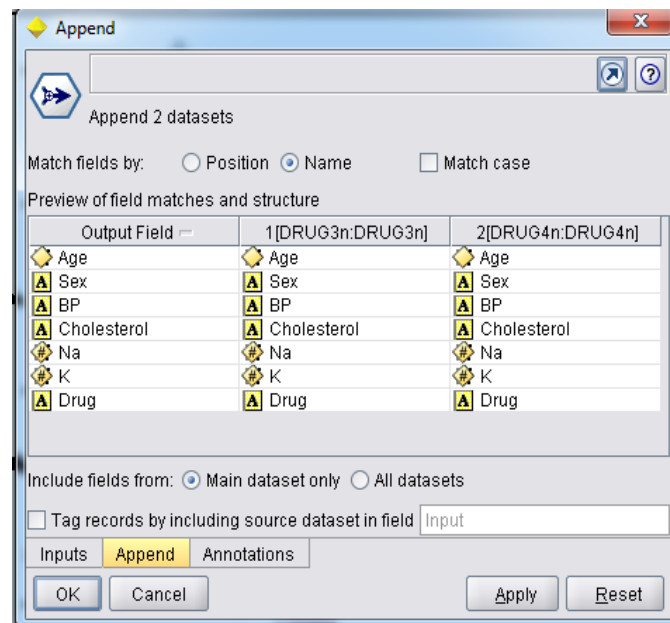
OK



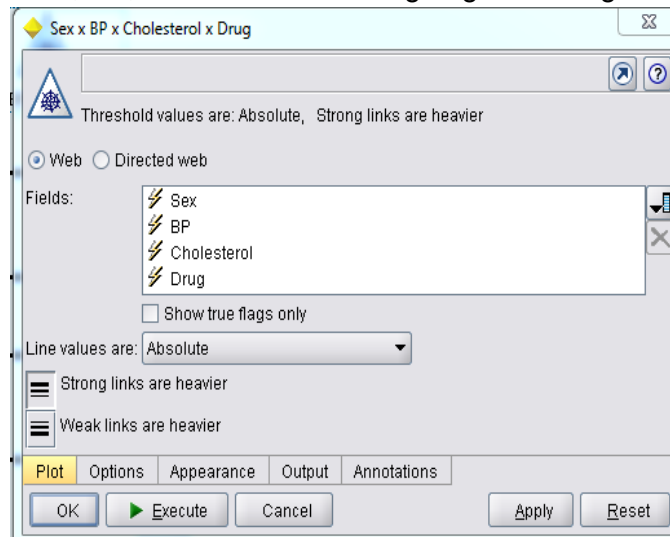
c) List 5 strong associations of attribute values, and derive and display the data.

Procedure: The following are the nodes used for this exercise with respective settings.

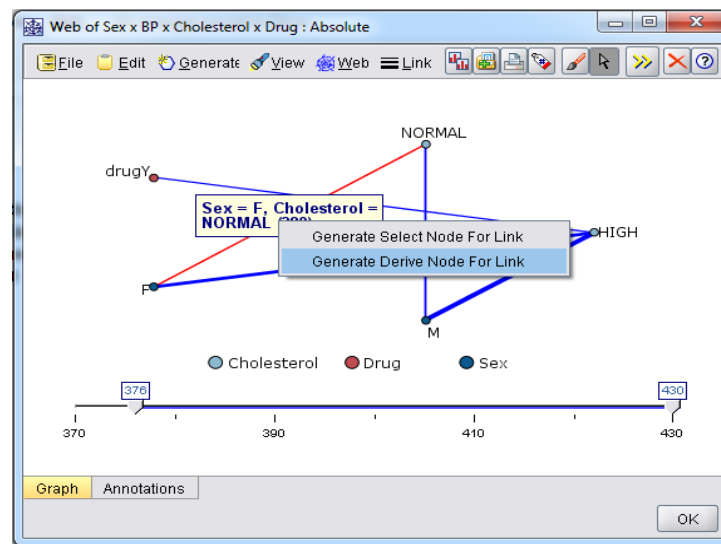
APPEND



WEB: Plotting the web for Sex, BP, Cholesterol and Drug to get 5 strong associations



When web is created showing 5 strong links, we have to derive nodes for every links by right clicking on link and generate derive node for link.

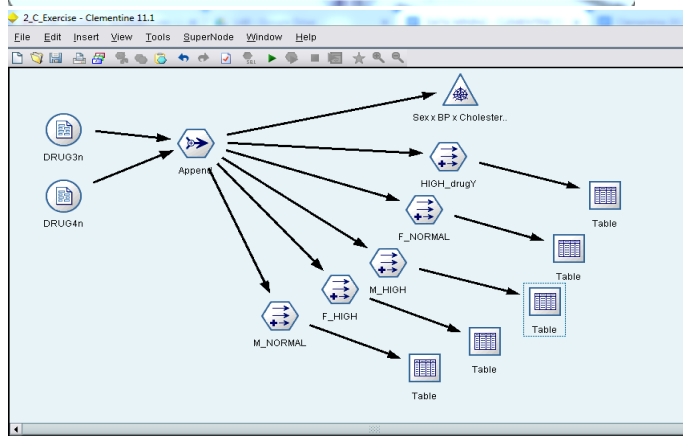


When a derive node is created from link the following is the configuration of derive node

OUTPUT:

Table (8 fields, 1,600 records) #5

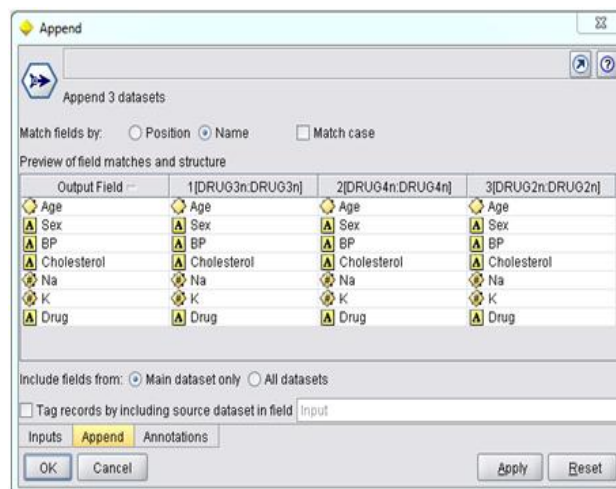
	Age	Sex	BP	Cholesterol	Na	K	Drug	M_HIGH
1	25	F	HIGH	HIGH	0.676	0.075	drugA	F
2	17	F	HIGH	HIGH	0.540	0.030	drugY	F
3	23	M	LOW	NORMAL	0.556	0.036	drugY	F
4	24	M	NORMAL	NORMAL	0.845	0.055	drugY	F
5	74	F	LOW	HIGH	0.850	0.077	drugC	F
6	40	F	NORMAL	HIGH	0.677	0.050	drugX	F
7	32	F	HIGH	HIGH	0.582	0.025	drugY	F
8	70	M	LOW	HIGH	0.716	0.037	drugY	T
9	64	M	HIGH	NORMAL	0.641	0.078	drugB	F
10	45	M	HIGH	HIGH	0.664	0.048	drugA	T
11	33	F	LOW	NORMAL	0.822	0.028	drugY	F
12	74	F	LOW	NORMAL	0.772	0.048	drugY	F
13	73	M	HIGH	NORMAL	0.792	0.062	drugB	F
14	38	F	LOW	HIGH	0.794	0.052	drugY	F
15	72	F	HIGH	NORMAL	0.534	0.021	drugY	F



d) Append DRUG2n dataset to given datasets and consider distinct values of age.

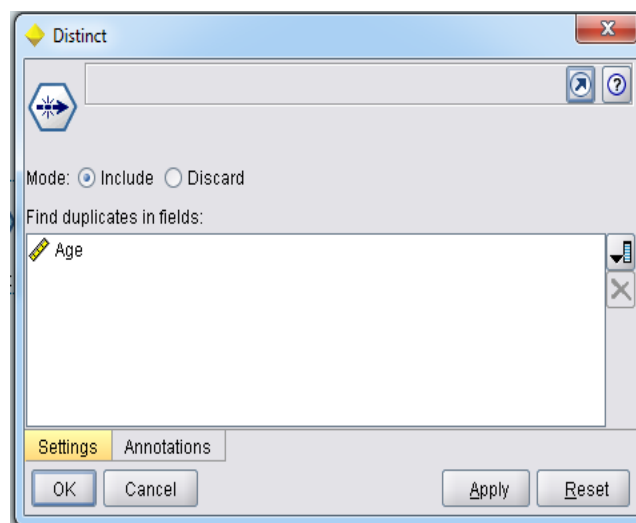
Procedure: The following are the nodes used for this exercise with respective settings.

Append:



Appended data sets are given in accordance with age are exported to output graphs.

Distinct Age:



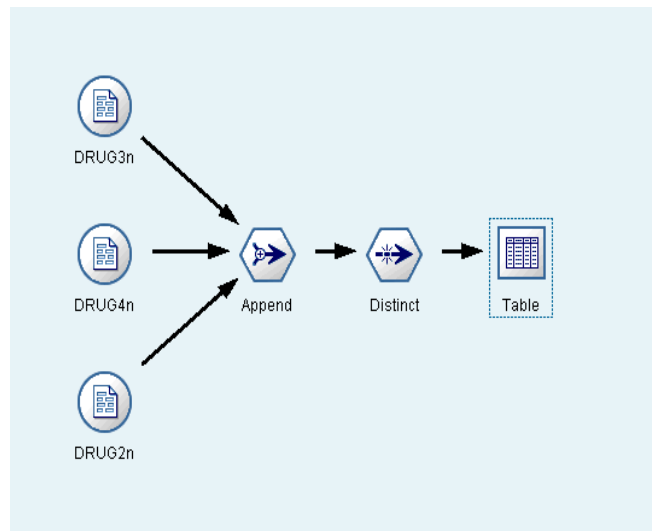
OUTPUT:

Table (7 fields, 60 records)

	Age	Sex	BP	Cholesterol	Na	K	Drug
1	25	F	HIGH	HIGH	0.676	0.075	drugA
2	17	F	HIGH	HIGH	0.540	0.030	drugY
3	23	M	LOW	NORMAL	0.556	0.036	drugY
4	24	M	NORMAL	NORMAL	0.845	0.055	drugY
5	74	F	LOW	HIGH	0.850	0.077	drugC
6	40	F	NORMAL	HIGH	0.677	0.050	drugX
7	32	F	HIGH	HIGH	0.582	0.025	drugY
8	70	M	LOW	HIGH	0.716	0.037	drugY
9	64	M	HIGH	NORMAL	0.641	0.078	drugB
10	45	M	HIGH	HIGH	0.664	0.048	drugA
11	33	F	LOW	NORMAL	0.822	0.028	drugY
12	73	M	HIGH	NORMAL	0.792	0.062	drugB
13	38	F	LOW	HIGH	0.794	0.052	drugY
14	72	F	HIGH	NORMAL	0.534	0.021	drugY
15	27	F	HIGH	NORMAL	0.555	0.047	drugA
16	62	M	HIGH	NORMAL	0.510	0.071	drugB
17	19	M	HIGH	NORMAL	0.553	0.033	drugY
18	28	M	HIGH	HIGH	0.584	0.068	drugA
19	54	M	NORMAL	NORMAL	0.606	0.077	drugX
20	37	F	NORMAL	HIGH	0.744	0.021	drugY

Table Annotations

OK



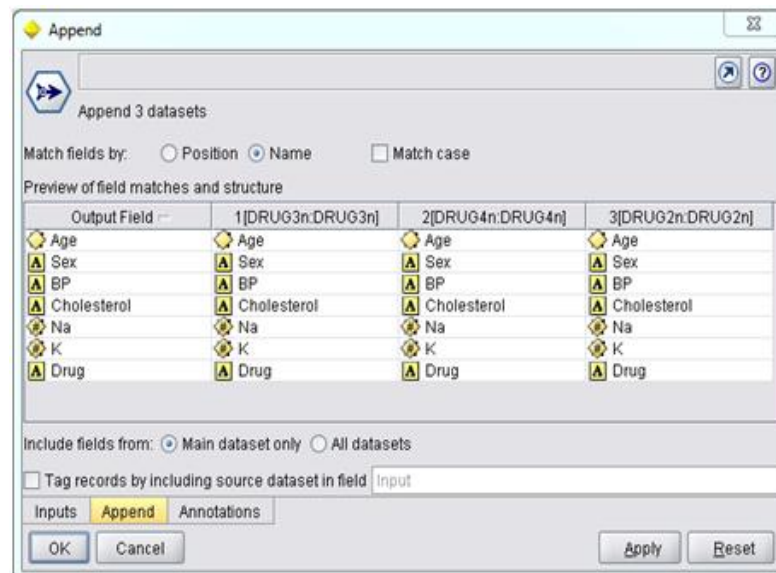
e) Using the above 3 datasets (DRUG2n, DRUG3n, DRUG4n) perform the following

i) Young_Age ≤ 30 , Middle_Age > 30 and ≤ 50 , Senior_Age > 50

ii) Multi plot the above Age categories with Na and K and drug

Procedure: The following are the nodes used for this exercise with respective settings.

Append:



Appended data sets are given in accordance with age are exported to output graphs.

Select Age = Young:

Young

Mode: ☒ Include ☐ Discard

Condition: Age <= 30

Settings Annotations

OK Cancel Apply Reset

Select Age = Middle:

Middle

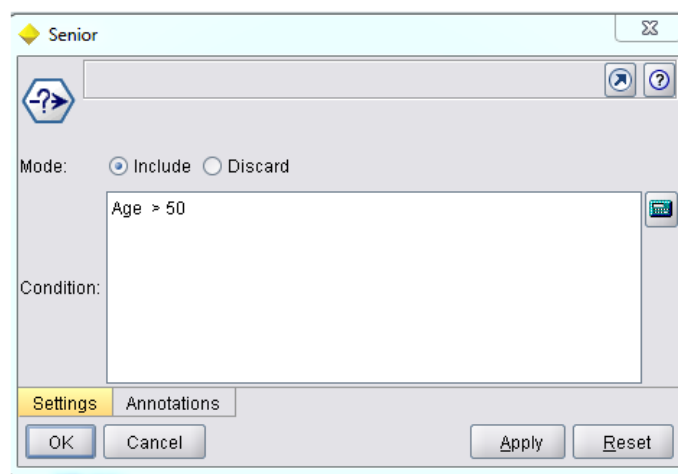
Mode: ☒ Include ☐ Discard

Condition: Age > 30 and Age <= 50

Settings Annotations

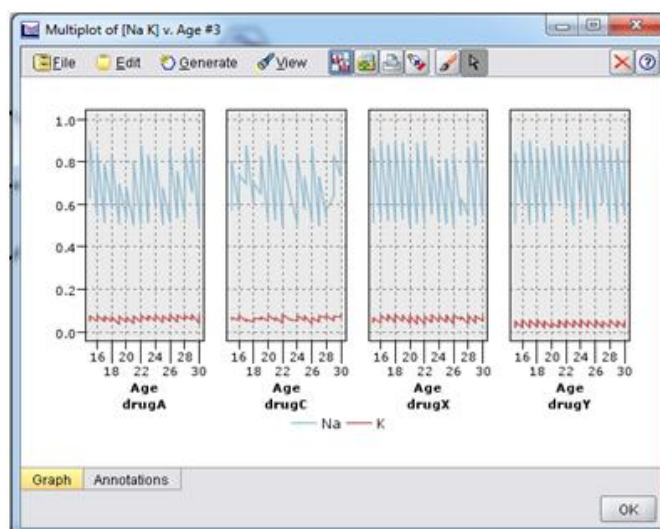
OK Cancel Apply Reset

Select Age = Senior:

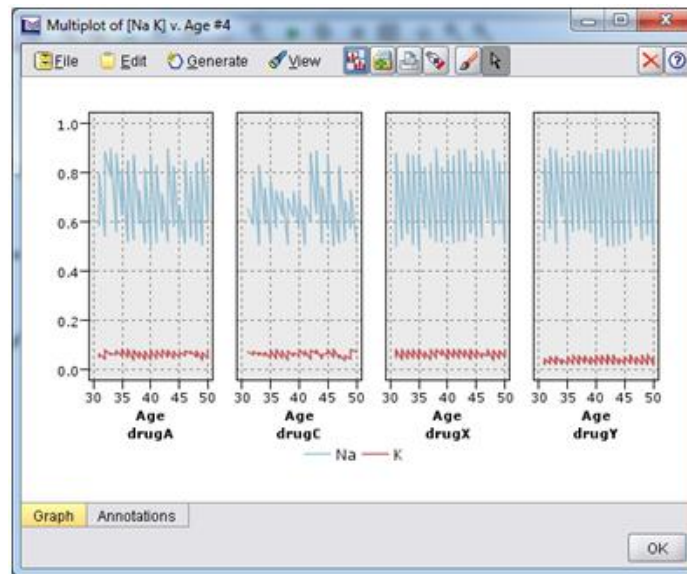


For each Select of age we use Multi Plot as shown in output.

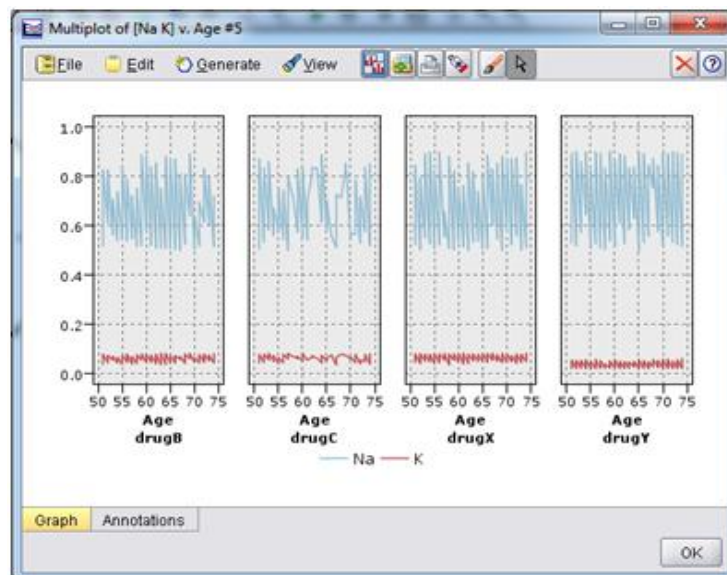
OUTPUT:



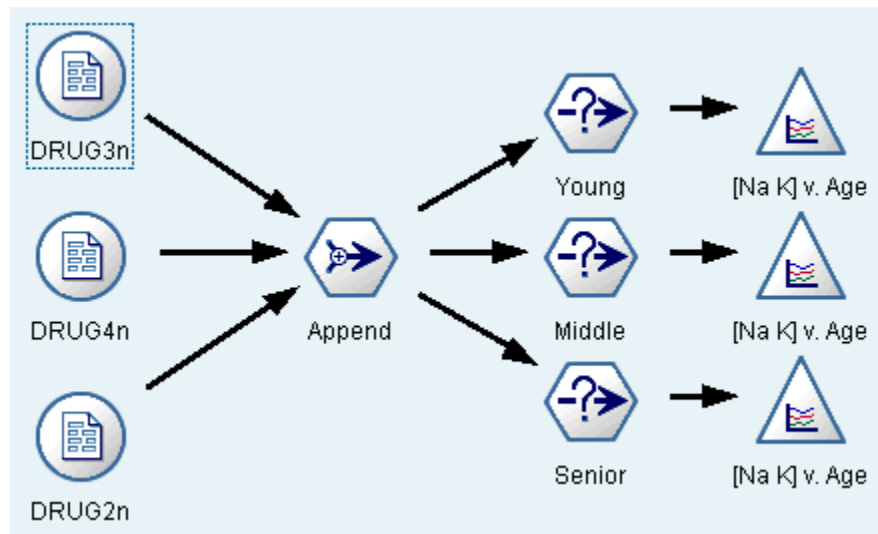
Above is Multi plot the above Young_Age categories with Na and K and drug



Above is Multi plot the above Middle_Age categories with Na and K and drug



Above is Multi plot the above Senior_Age categories with Na and K and drug



WEKA TOOL

Demonstrate the Weka machine learning toolkit and perform the following exercises

- 1. a. Determine the numbers of instances of each class are present in the data Load the weather dataset and**

Open Weka:

- Launch the Weka Explorer.

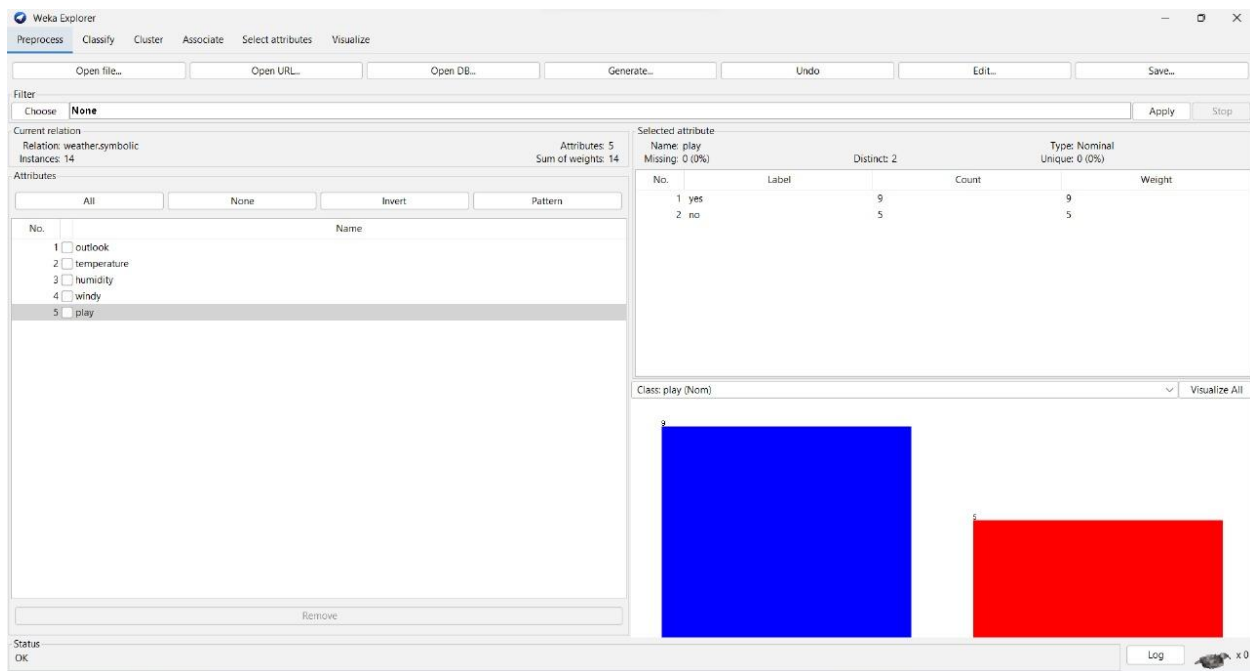
Load the Weather Dataset:

- Go to the "Preprocess" tab.
- Click on "Open file" and select the weather dataset.

Determine Instances by Class:

- In preprocess tab itself by clicking play attribute we get the output as number of instances in the form of Bar graph.
- So, the output is mentioned below.

OUTPUT:



b. remove all instances where the attribute 'humidity' has the value 'high'?

Load the Weather Dataset:

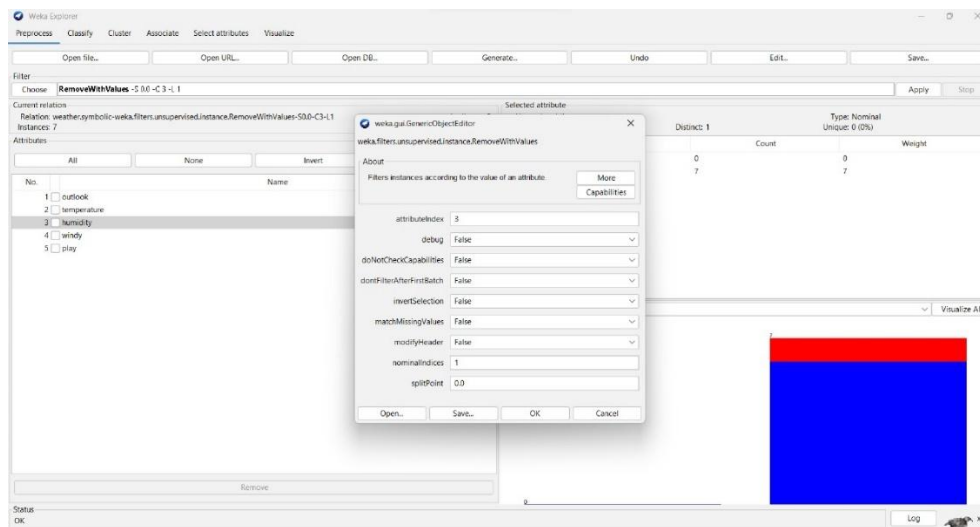
- Click on the "Open file" button in the "Preprocess" tab.
- Select the weather dataset file and click "Open."

Go to the "Preprocess" Tab:

- Click on the "Preprocess" tab at the top of the Weka Explorer.

Use the "RemoveWithValues" Filter:

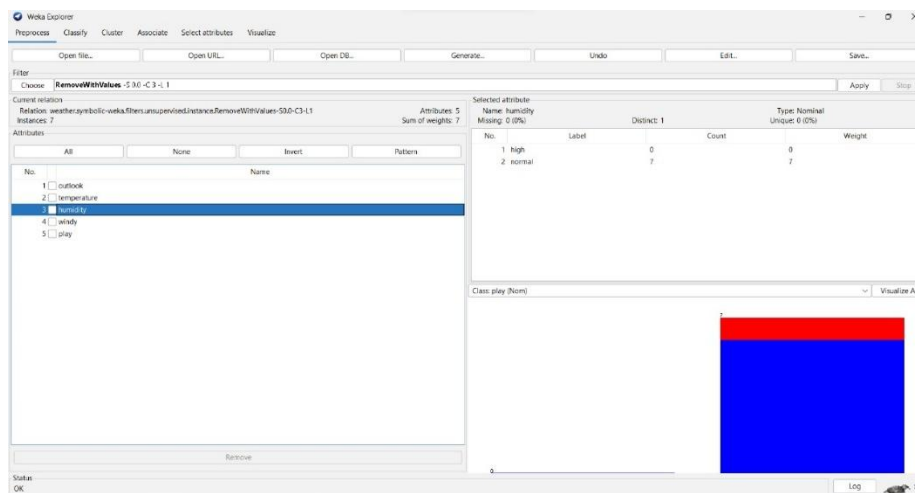
- In the "filters" panel on the left, look for the "unsupervised.instance.RemoveWithValues" filter.
- Double-click on it to add it to the list of applied filters on the right.



Configure the "RemoveWithValues" Filter:

- In the "attributeName" field, specify the name of the attribute you want to filter on ('humidity').
- In the "nominalIndices" field, specify the index of the value to be removed. Since 'high' is likely to be the first nominal value, you can set it to "1" (assuming indices start from 1).
- Click the "Apply" button to execute the filter.

OUTPUT:



2. Load the iris dataset and view the Visualizer panel.

Load the Iris Dataset:

- In the "Preprocess" tab, click on "Open file" and select the iris dataset.

Visualize with the Visualizer Panel:

- Go to the "Visualize" tab.
- We can use various visualization tools, such as the "Visualize" button to see attribute distributions, the "Classify" button for classification, or the "Cluster" button for clustering.

Explore Visualizations:

- Click on different attributes to visualize their distributions.
- Utilize the scatter plot to observe relationships between attributes.

OUTPUT:

